

Arduino Smart Street Light using LDR

Objective: Automatically turn an LED (street light) ON in darkness and OFF in daylight using an LDR sensor.

Arduino Code

```
const int ldrPin = A0;
const int ledPin = 13;

void setup() {
  pinMode(ledPin, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int ldrValue = analogRead(ldrPin);
  Serial.println(ldrValue);

  if (ldrValue < 500) {
    digitalWrite(ledPin, HIGH);
  } else {
    digitalWrite(ledPin, LOW);
  }

  delay(500);
}
```

Explanation

- ldrPin (A0): Reads the light intensity from the LDR sensor.
- ledPin (13): Controls the LED that represents the street light.
- setup(): Configures the LED pin as OUTPUT and starts Serial communication.
- analogRead(): Reads the LDR value (0-1023).
- if (ldrValue < 500): When it is dark, the LED turns ON.
- else: When there is enough light, the LED turns OFF.
- delay(500): Waits for 500 milliseconds before repeating.

Expected Output

Dark condition: LED ON

Bright condition: LED OFF